IT PROJECT MANAGEMENT · CHAPTER 10

Communication Management

Most of a project manager's time is spent communicating. This chapter plans who needs what information, when, in what form, and how to make sure it arrives and is understood.

PMBOK Guide 7th ed. **Scrum Guide 2020** **Case study: Realtime Quiz**

AGENDA

What we will cover

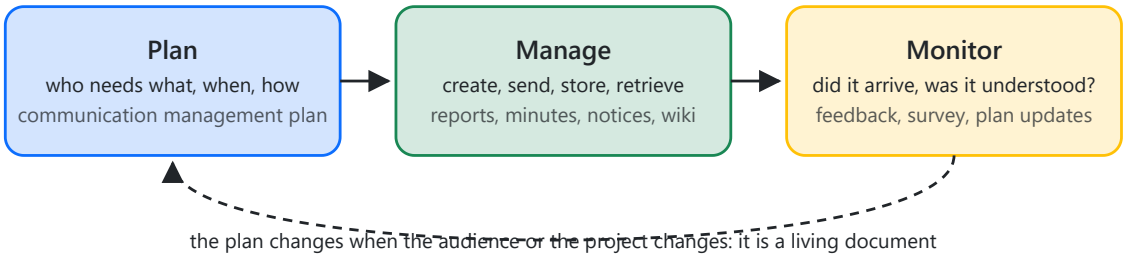
1. Communication models, channels and the number of communication paths
2. Communication requirements analysis by stakeholder
3. Communication management plan: purpose, audience, frequency, medium, owner
4. Meetings that work: agendas, minutes, decisions and actions
5. Status and progress reporting: dashboards, earned value reports, burndown charts
6. Written communication for IT projects: status emails, decision records, release notes
7. Tools: chat, wikis, issue trackers, video calls, and their appropriate use
8. Communicating bad news, escalation and conflict over information
9. Information distribution, retention and confidentiality
10. Cross-cultural and remote communication
11. Chapter Quiz
12. Lab-10

Click any item to jump directly to that topic.

Why communication is most of the job

- A widely quoted PMI figure: project managers spend about **90 % of their time communicating**: asking, explaining, reporting, negotiating, writing.
- Chapters 04 to 09 produced plans and baselines. None of them helps if the sponsor never reads the status, the team does not know the decision, or the pilot students learn about an outage from a blank screen.
- The **PMBOK Guide** describes three activities: *plan* communications, *manage* them (create, distribute, store) and *monitor* whether the information needs were actually met. PMBOK 7 folds them into the Stakeholder and Team performance domains.
- Output of this chapter and Lab-10: a **communication management plan**, meeting and report templates, and the ability to write bad news well.

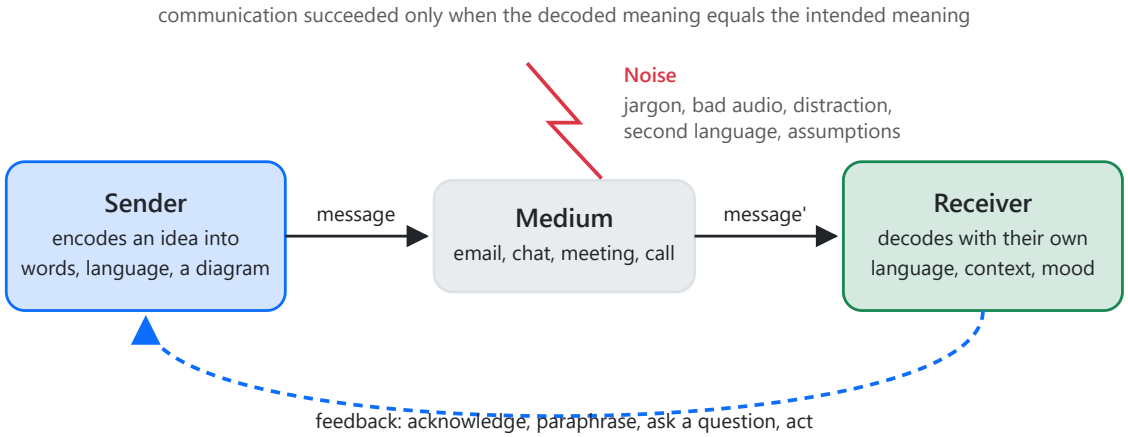
inputs from earlier chapters: stakeholder register (Ch. 01, 11), schedule (Ch. 05), EVM baseline (Ch. 06), risk register (Ch. 09)



PMBOK Guide 6th ed. processes; PMBOK Guide 7th ed. performance domains

TOPIC 1

The sender-receiver model: where messages break



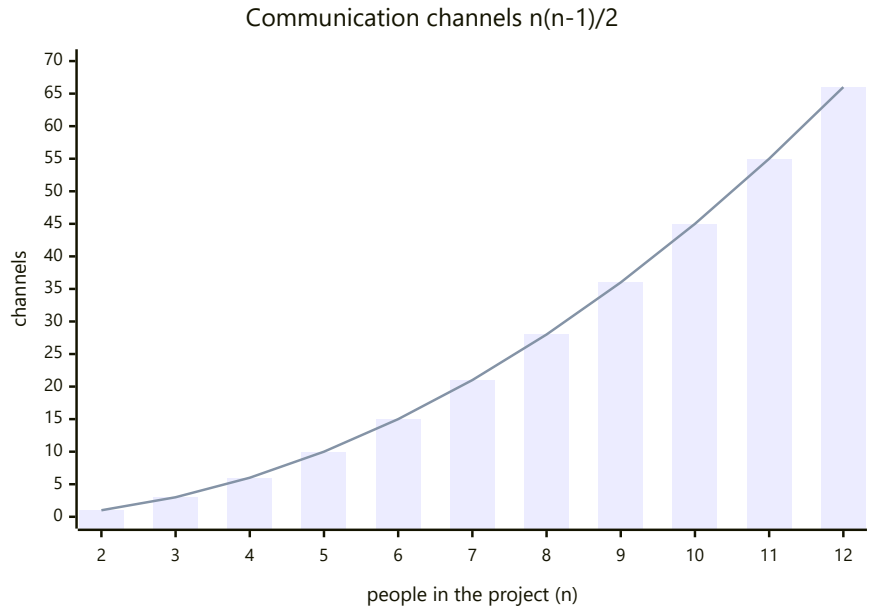
"Received" is not "understood". Only feedback closes the loop.

basic sender-receiver model as summarised in the PMBOK Guide

- **Encode**: the sender turns an idea into a message. The choice of words, language (Khmer or English), level of detail and diagram is already a decision.
- **Medium**: the channel carrying the message. Each medium loses something: chat loses tone, email loses immediacy, a call loses the record.
- **Noise**: anything that distorts the message on the way: technical (bad Wi-Fi), semantic (jargon, acronyms), cultural (indirect "yes"), personal (stress, multitasking).
- **Decode**: the receiver reconstructs meaning from their own context. The same sentence means different things to a sponsor and a developer.
- **Feedback**: the receiver's response proves what was understood. Without it the sender only knows the message was *sent*.

Practical test: after an important message, ask the receiver to say back what they will do. "Deploy is done" can mean "on my laptop" or "on gicitc.com".

Communication channels grow with the square of the team




```
Channels = n (n - 1) / 2

n = 4     4 x 3 / 2   =  6
n = 6     6 x 5 / 2   = 15
n = 10   10 x 9 / 2   = 45

Realtime Quiz:
4 students + PM        n = 5  -> 10
+ sponsor + hosting    n = 7  -> 21
+ 2 pilot instructors  n = 9  -> 36

Adding 1 person to a team of 6
adds 6 channels (15 -> 21), not 1.
```

- Every channel is a place where the message can be distorted or simply not sent.
- Therefore: keep the core team small, route information through **agreed hubs** (a status report, a board, a wiki page) instead of 36 private chats, and define who talks to the sponsor and the hosting provider.
- Quiz alert: the question usually says "the team grows from 4 to 6: how many *new* channels?" Answer: 15 - 6 = 9.

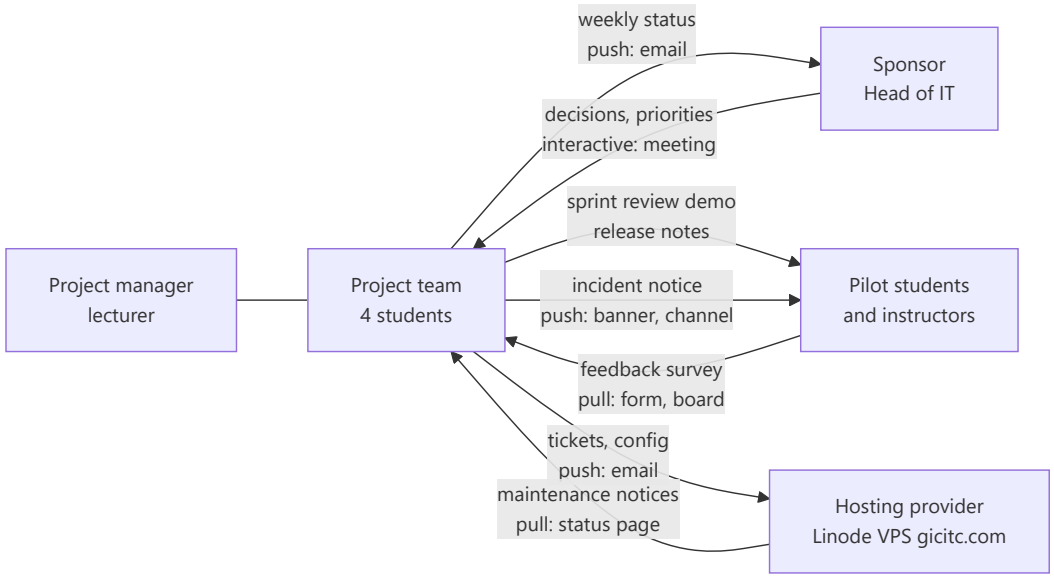
Communication requirements analysis: who needs what, and why

Stakeholder (Lab-01)	Information needed	Why	How often	Format	Sensitivity
Sponsor: Head of IT	Progress vs baseline, RAG, top risks, decisions needed	Funds hosting, approves go-live	Weekly	One-page status email	Internal
Project manager (lecturer)	Task state, blockers, hours spent, defects	Steers the team, grades effort	Daily	Board, stand-up, timesheet	Internal
Team (4 students)	Sprint goal, decisions, who does what	Do the work without waiting	Daily	Stand-up, chat, issue tracker	Internal
Pilot instructors (2)	Features ready, how to author a quiz, known issues	Prepare quizzes for the pilot	Per sprint	Demo, release notes	Public
Pilot students (50)	How to join, changes, incidents	Adopt the tool, trust it	Per release, on incident	Announcement, banner	Public
Hosting provider (Linode, gicitc.com)	Resource needs, tickets, maintenance windows	Keep the VPS running	Monthly, on demand	Support ticket, email	Confidential (credentials)
Campus network admin	Ports and proxy rules for WebSocket	Students must be able to connect	Once, on change	Memo with technical annex	Internal
Faculty management	Milestones, go-live date, budget use	Governance, escalations	Monthly	Summary paragraph	Internal

- Start from the **stakeholder register** of Lab-01 and the org chart of Lab-08. Every stakeholder gets one row; if a row has no "why", nobody needs that information.
- Ask each stakeholder five questions: *What decisions do you make about this project? What do you need to know to make them? How often? In which format? What must you never receive by accident?*
- **Sensitivity** decides the channel: credentials never travel in the group chat, student names never appear in a public release note.
- Frequency follows the decision cycle of the receiver, not the convenience of the sender: the sponsor decides weekly, the team decides daily.

⚠ Over-communication is also a failure: a sponsor who receives 40 GitHub notifications a day stops reading the one email that matters.

Information flow: push, pull and interactive PMBOK Guide



Method	Meaning	Use when
Push	Sender delivers to specific receivers; delivery is certain, understanding is not	Status reports, memos, incident notices, release emails
Pull	Receivers fetch when they need it	Wiki, repository, dashboard, recorded demo, large audiences
Interactive	Two or more parties exchange in real time; best for understanding	Stand-ups, reviews, negotiations, bad news, conflicts

- Draw the flow once and you see the gaps: who tells the campus network admin about the WebSocket port? Who hears the hosting provider's maintenance notice at 02:00?
- Each arrow later becomes one row of the communication management plan.

The communication management plan: one row per flow

#	Audience	Purpose	Content	Method	Frequency	Owner	Feedback loop
C1	Sponsor (Head of IT)	Weekly status	RAG per area, EVM (SPI, CPI, EAC), top 3 risks, decisions needed	Email, one page, push	Friday 16:00	PM	Sponsor replies or comments by Monday noon
C2	Team + PM	Daily stand-up	Yesterday, today, blockers; board walk	In person or Google Meet, 15 min, interactive	Daily 09:00	Rotating facilitator	Board updated before the call; blockers become issues
C3	Pilot instructors, sample of students	Sprint review	Demo of the increment, what changed, what is next	Meeting + live demo on gicitc.com, interactive	Every 2nd Friday	Team (rotating presenter)	Feedback form, items added to backlog
C4	Pilot students (50)	Incident notice	What is broken, impact, workaround, next update time	Banner in the app + Telegram channel, push	Within 30 min of a P1 incident, then hourly	On-call developer	Closing notice with cause; questions answered in channel
C5	Hosting provider	Operations	Resource requests, outage reports, maintenance confirmation	Support ticket, push	On demand; monthly check	DevOps student (Vanna)	Ticket closed with resolution copied to wiki
C6	Faculty management	Governance summary	Milestones reached, next milestone, budget use, escalations	Email paragraph, push	Monthly, 1st Monday	Sponsor	None required; escalations answered in 3 days
C7	Team + PM	Retrospective	What went well, what to change, 1 to 3 improvement actions	Meeting, 45 min, interactive	End of each sprint	PM	Actions tracked as issues, reviewed next retro
C8	Everyone	Single source of truth	Plan, decisions, minutes, release notes	GitHub wiki + repo, pull	Continuous	PM (structure), authors (content)	Broken or stale page reported as issue

- **Audience**: a stakeholder or group from the register, never "all".
- **Purpose**: the decision or action it supports.
- **Content**: the fixed sections, so every issue looks the same.
- **Method**: push, pull or interactive plus the tool.
- **Frequency**: a day and time, or a trigger ("within 30 min of a P1").
- **Owner**: one name; a role is fine if the roster is in the plan.
- **Feedback loop**: how you learn it was received and understood.

☑ Test each row: could a new team member execute it from the row alone?

The plan as a document: `communication-management-plan.md`

```
# Communication Management Plan - Realtime Quiz
Version 1.0 | Owner: project manager | Approved by: sponsor, week 3

## 1. Purpose and scope
Who receives which information about the project, when and how.
Covers the team, sponsor, pilot users, hosting provider, campus IT.

## 2. Stakeholder communication requirements
See `communication-requirements.csv` (one row per stakeholder).

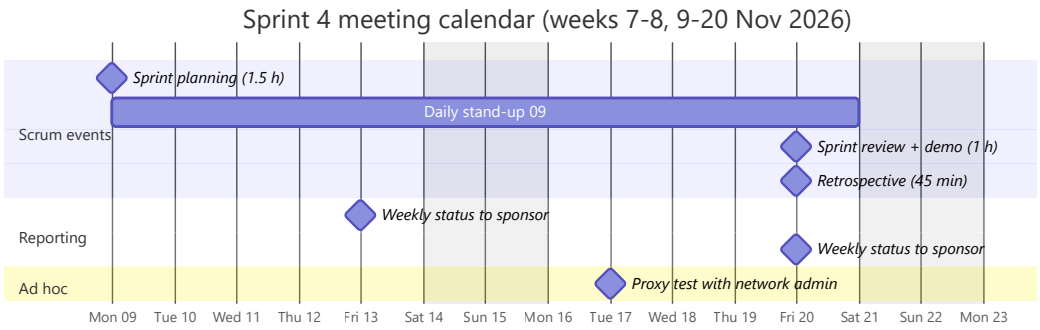
## 3. Communication matrix
See `communication-plan.csv`: id, audience, purpose, content,
method, frequency, owner, feedback loop (rows C1..C8).

## 4. Meetings          -> agendas, minutes, ground rules (section 4)
## 5. Reporting         -> status report template, RAG thresholds
## 6. Tools             -> which tool for what, response times
## 7. Escalation path   -> levels, time limits, who decides
## 8. Information store -> wiki structure, naming, retention, classes
## 9. Glossary          -> RAG, SPI, CPI, EAC, P1, MVP, sprint
## 10. Change history   -> plan changes are communicated too
```

Section	Answers
Requirements	Who needs what and why (Topic 2)
Matrix	Purpose, audience, content, method, frequency, owner
Meetings	Which meetings exist, their timebox and outputs
Reporting	Report format, thresholds for Green, Amber, Red
Escalation	When a problem leaves the team and who decides
Information store	Where documents live, naming, retention, confidentiality
Glossary	Shared vocabulary; add Khmer terms where useful

- The PMBOK Guide lists exactly these components: requirements, information to be communicated (language, format, content, level of detail), reason, time frame, responsible person, methods, resources, escalation process, update method, glossary, flowcharts and constraints.
- Keep the matrix in CSV next to the Markdown so it can be sorted and filtered; the Markdown explains, the CSV executes.

A meeting system, not meetings: the sprint calendar Scrum Guide 2020



Meeting	Timebox (2-week sprint)	Output
Sprint planning	1.5 h (Scrum Guide: max 8 h per month-long sprint)	Sprint goal, sprint backlog
Daily stand-up	15 min, same place, same time	Updated board, blockers as issues
Sprint review	1 h (max 4 h per month)	Feedback, backlog changes
Retrospective	45 min (max 3 h per month)	1 to 3 improvement actions
Sponsor status	Written, 15 min call if Amber or Red	Decisions, approvals

```
Meeting cost = participants x duration
Stand-up: 5 x 0.25 h x 5 days x 14 wk = 87.5 h
          = 14.6 % of BAC (600 h)
Drift to 30 min: +87.5 h = one student's
whole sprint. Timebox and start on time.
```

Agenda before, minutes after: decisions and actions are the product

```
# Sprint 4 review - minutes
Date: Fri 20 Nov 2026, 14:00-15:00 | Room B204 + Meet
Present: PM, Dara, Sokha, Vanna, Chan, 2 pilot instructors
Absent: - | Minutes: Chan | Facilitator: Sokha

## Agenda (sent Wed 18 Nov)
1. Sprint goal check: scoring + leaderboard demo          (25 min)
2. Feedback from instructors                              (15 min)
3. Timer drift seen in the demo on Wi-Fi                  (10 min)
4. Next sprint preview and pilot readiness                (10 min)

## Decisions
| Id   | Decision                                    | Made by     |
|------|---------------------------------------------|-------------|
| D-12 | Leaderboard shows top 10 only, not all 50   | instructors |
| D-13 | Freeze "Could" features until MVP is demoed | PM          |

## Actions
| Id   | Action                                      | Owner | Due    |
|------|---------------------------------------------|-------|--------|
| A-31 | Fix timer drift #142 on campus Wi-Fi        | Sokha | 25 Nov |
| A-32 | Send instructor quiz-authoring guide v0.4   | Dara  | 23 Nov |
| A-33 | Book proxy test with network admin          | Vanna | 24 Nov |

## Parking lot (not decided today)
- Anonymous join with nickname: instructors want it, needs sponsor OK.
```

- **Agenda** 48 hours before: purpose, items with owner and minutes, pre-reading link. No agenda, no meeting.
- **Minutes** within 24 hours, in the wiki, linked from the chat. Not a transcript: attendance, decisions, actions, parking lot.
- **Decision** = a sentence a stranger could apply, plus who made it. Numbered (D-12) so later documents can refer to it.
- **Action** = verb, owner (one name), due date. Copy each into the issue tracker; the minutes are the record, the tracker is the reminder.
- **Ground rules** the team agrees in Sprint 1: start on time, cameras on when remote, one conversation at a time, phones away, disagree in the meeting not after it, Khmer or English but every decision written in English.

⊗ Anti-pattern: "We discussed the leaderboard." Nobody knows what was decided, and the same discussion happens next week.

Status reporting: one page the sponsor reads in two minutes

Realtime Quiz · status week 8 of 14 · Fri 20 Nov 2026 — Amber

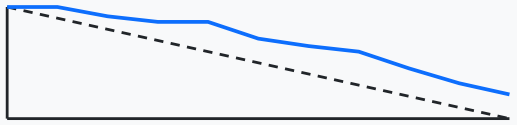
Scope	Schedule	Cost (hours)	Risks	Quality
MVP unchanged	SPI 0.90	CPI 0.85	R-03 test booked	0 open P1 defects

Earned value (person-hours, BAC 600)

PV 340	EV 306	AC 360
SV -34	CV -54	EAC 706

Cash: 65 of 115 USD spent (VPS, domain)

Sprint 4 burndown · day 10: 9 of 40 pts left



Top risks and decisions needed

R-03 WebSocket blocked by campus proxy: test booked Tue 24 Nov with network admin (Vanna)

R-05 Scope creep from "Could" features: frozen by decision D-13 until the MVP demo

Decision needed by Mon 23 Nov: approve a one-week buffer before the pilot? (Sponsor)

- **RAG** (Red, Amber, Green) per area with *written thresholds*, otherwise every report is green until the disaster: Green SPI and CPI ≥ 0.95; Amber 0.85 to 0.94; Red < 0.85 or a milestone missed.
- **Overall status** is the worst area, never the average. Risks: Amber while a high risk is open with a booked response, Red when it materialises or has no response.
- **Earned value** (Chapter 06) gives the sponsor numbers instead of adjectives: "10 % behind and 15 % over on hours" beats "slightly late".
- **Trend** matters more than the snapshot: show last week's values or a burndown, so the reader sees direction.
- **Decisions needed** with a date and a name turn a report into a tool; a report without an ask is easy to ignore.

Dashboards are *pull*: the sponsor also gets the same numbers *pushed* in the Friday email (row C1).

Worked example: the week 8 status report with Lab-06 numbers

```
STATUS REPORT - Realtime Quiz - week 8 of 14 (16-20 Nov 2026)
Overall: AMBER (was GREEN)      To: Head of IT   From: PM

1. Headline
   Scoring and leaderboard demoed (sprint 4 goal met). We are 10 %
   behind plan and 15 % over on hours; pilot date at risk.

2. Progress this week          3. Plan next week
   - Leaderboard broadcast done    - Results API and CSV export
   - Reconnect and resume done     - Proxy test on campus Wi-Fi
   - 9 of 40 points carried over   - Sprint 5 planning Mon 09:00

4. Earned value (person-hours)   PV 340  EV 306  AC 360  BAC 600
   SV -34 h   CV -54 h   SPI 0.90   CPI 0.85   EAC 706 h   VAC -106 h
   Cash: 65 USD of 115 USD spent (VPS 3 months, domain).

5. RAG   Scope G   Schedule A   Cost A   Risks A   Quality G

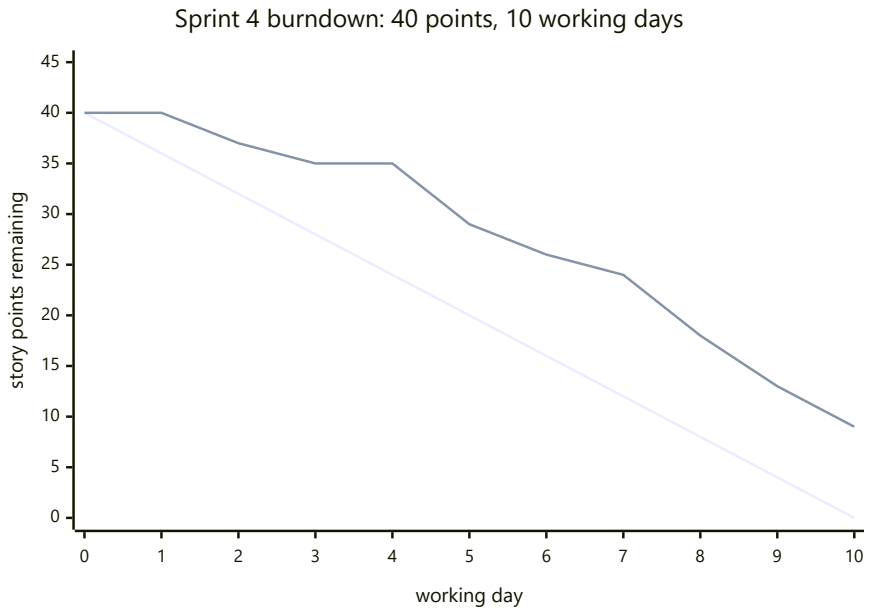
6. Top risks
   R-03 Proxy blocks WebSocket   P 0.4  I high   test 24 Nov (Vanna)
   R-05 Scope creep              P 0.3  I med    frozen by D-13
   R-08 Exam week 12 absence     P 0.5  I med    plan buffer

7. Decisions needed
   By Mon 23 Nov: approve one-week buffer before pilot? (Sponsor)
```

Measure	Formula	Week 8
SV	EV - PV	306 - 340 = **-34 h**
CV	EV - AC	306 - 360 = **-54 h**
SPI	EV / PV	306 / 340 = **0.90**
CPI	EV / AC	306 / 360 = **0.85**
EAC	BAC / CPI	600 / 0.85 = **706 h**
ETC	EAC - AC	706 - 360 = **346 h**
VAC	BAC - EAC	600 - 706 = **-106 h**

- Same seven sections every week, so the reader finds section 7 without searching.
- The headline states the change since last week ("was GREEN") and the consequence ("pilot date at risk"), not the activity.
- Risks quoted from the Lab-09 register by id, with the response and its owner.
- Fits on one page. Details live in the wiki and are linked, not pasted.

Burndown chart: the team's own progress picture




```
xychart-beta
  title "Sprint 4 burndown"
  x-axis "working day" [0,1,2,3,4,5,6,7,8,9,10]
  y-axis "points remaining" 0 --> 45
  line [40,36,32,28,24,20,16,12,8,4,0]
  line [40,40,37,35,35,29,26,24,18,13,9]
```

- First line: **ideal** burn, 40 points over 10 days = 4 per day. Second line: **actual** remaining points at the end of each day.
- Reading: flat on days 1 and 4 (nothing finished: large stories or blocked), a step on day 5 (a 6-point story closed). Day 10 ends at 9 points: carried over to Sprint 5 and mentioned in the status report.
- Burndown is for the team and the review; the sponsor gets EVM. Same truth, different audience.
- Story points count only when the story meets the Definition of Done, otherwise the chart lies.

Written communication: the status email

Rule	Why
Subject = project, period, status	Searchable, readable in a phone notification
BLUF: bottom line up front	The reader may stop after two lines
Numbers, not adjectives	"10 % behind" can be acted on, "a bit late" cannot
One clear **ask** with a deadline	The reader knows what you need from them
Link to detail, do not paste it	Keeps the email under 200 words
Same structure every week	Reading becomes a 2-minute habit

💡 Write the ask first, then the headline, then the rest. If you cannot name an ask, the email is information only: say so ("no action needed").

```
Subject: Realtime Quiz - week 8 status - AMBER (was GREEN)

Dear Mr. Sovann,

Bottom line: sprint 4 goal met (scoring, leaderboard), but we are
10 % behind schedule (SPI 0.90) and 15 % over on hours (CPI 0.85).
The week-14 pilot is at risk unless we add a one-week buffer.

Ask: please confirm by Monday 23 Nov whether the pilot may move to
week 15 if the proxy test on 24 Nov fails. Details in section 7.

This week: leaderboard broadcast and reconnect done; 9 of 40 points
carried over. Next week: results API, CSV export, proxy test.

Top risk: R-03 campus proxy may block WebSocket (P 0.4, impact high).
Test booked with the network admin; fallback is SockJS (2 days).

Full report and dashboard: wiki/reports/2026-11-20-status-week-08.md

Best regards,
Sreymom (project manager)
```

Decision records and release notes: writing for the future reader

```
# D-012 Leaderboard shows the top 10 only
Date: 2026-11-20 | Status: accepted | Owner: PM
Made in: Sprint 4 review | Supersedes: -

## Context
Instructors saw 50 rows scroll on the projector;
students at the bottom felt exposed (feedback form,
7 of 12 comments).

## Options considered
1. Show all participants (current)
2. Show top 10, own rank always visible  <- chosen
3. Instructor toggle per session (more work, later)

## Decision
Leaderboard broadcasts the top 10 plus the viewer's
own rank and score. Option 3 goes to the backlog.

## Consequences
- LeaderboardService change, issue #148, 3 points
- Release note v0.5.0; instructor guide updated
- Revisit if a class of fewer than 15 students asks
```

```
# Realtime Quiz v0.4.0 - Sprint 4 (20 Nov 2026)
Audience: pilot instructors and students. Deployed on
https://quiz.gicitc.com on 20 Nov 18:00, downtime 5 min.

## New
- Live leaderboard after every question
- Scores saved: a closed session can be reviewed later
- Rejoin a session after losing the connection

## Changed
- Timer now follows the server clock (less drift)

## Fixed
- Late answers no longer counted after the timer (#131)

## Known issues
- Leaderboard shows all participants; top 10 comes in
  v0.5.0 (decision D-012)
- Campus Wi-Fi may block the live connection; if the
  page stays on "connecting", switch to mobile data

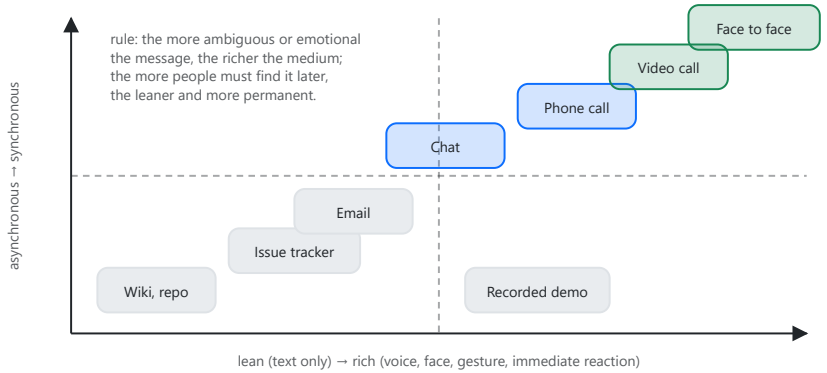
## Need help?
Telegram channel @rtquiz-pilot or quiz@gicitc.com
```

☑ **Decision record**: context, options, decision, consequences. Six months later nobody remembers *why*; the record does. Software teams call the same thing an ADR.

☑ **Release note**: written for users, not developers: what they can do now, what changed for them, what still hurts, where to get help. No class names, no commit hashes.

Tools: match the medium to the message

Method	Tool (Realtime Quiz team)	Good for	Bad for	Response expected
Interactive	Stand-up, review (room or Google Meet)	Decisions, feedback, bad news, conflict	Information that could be read	Immediate
	Chat (Telegram group)	Quick questions, coordination, links	Decisions (they get lost), long arguments, credentials	Within the working day
Push	Email	Status to sponsor, memos to campus IT, anything formal	Fast coordination, discussions	1 to 2 working days
	Announcement banner, Telegram channel	Incident notices, release notes to 50 students	Two-way discussion	None; questions come by chat
Pull	GitHub Issues + Projects board	Tasks, bugs, actions from minutes, burndown	Discussions unrelated to a work item	Assignee updates at stand-up
	GitHub wiki + repository	Plan, decisions, minutes, reports: the single source of truth	Anything that changes hourly	None; pages have owners
	Dashboard, recorded demo	Sponsor and faculty self-service; absent students	Anything needing acknowledgement	None



TOPIC 7

Tool rules: agree them once, in Sprint 1

- **Chat is for now, the tracker is for later.** Any request that takes more than 15 minutes of work becomes an issue with an assignee. "Can someone look at the timer?" in chat is not a task.
- **Decisions leave the chat.** A decision made in chat is copied into a decision record or the minutes within a day, otherwise it did not happen.
- **One channel per audience.** Team group (internal), pilot channel (public announcements, read-only), sponsor by email. Never mix: a student screenshot in the team group must not contain credentials.
- **Video calls** for anything with disagreement or emotion; camera on; one person writes the minutes live in the shared document.
- **Notifications**: everyone watches the repository, nobody is expected to answer chat after 21:00 or before 08:00; a P1 incident uses a phone call, not a message.
- **Secrets** (JWT secret, database password, Linode token) live only in the password manager and the server environment; sharing one in chat triggers a rotation.

```
## Issue #142 - Timer ends 2-3 s early on campus Wi-Fi
Labels: bug, P2, sprint-5     Assignee: Sokha     Points: 3

### Observed
In the Sprint 4 review (20 Nov, room B204) the student
countdown reached 0 while the host console still showed 2 s.
Reproduced 3 of 5 times on campus Wi-Fi, 0 of 5 on 4G.

### Expected
Client drift under 200 ms after clock-offset correction
(NFR "Timer accuracy" in the project plan).

### Suspected cause
Offset measured once at connect; Wi-Fi latency spikes
(120-400 ms) skew the first sample.

### Acceptance
- Offset uses the median of 5 samples
- Drift test on campus Wi-Fi passes 10 of 10 (Vanna)
- Mentioned in release notes v0.5.0

Source: minutes 2026-11-20 sprint-4-review, action A-31
```

Bad news: the pilot slips one week. Two emails, same facts

⊗ Badly: buried, vague, blaming, no ask

```
Subject: Update

Hi,

Hope you are well. The team has been working very hard
this sprint and we made great progress on the export
feature and the leaderboard, which the instructors
really liked. There were some issues with the network
that were not our fault because the campus proxy
blocks WebSocket, and the network admin was not
available, so some things took longer than expected.
We might need a bit more time for the pilot, maybe
around one week or so, but we are confident it will
be fine. Let me know what you think.

Thanks,
Team
```

✓ Well: headline, impact, cause, plan, ask

```
Subject: Realtime Quiz - pilot moves from week 14 to
         week 15 - decision needed by Fri 18 Dec

Dear Mr. Sovann,

The pilot with 50 students cannot start in week 14 as
planned. I propose week 15 (Wed 6 Jan instead of 30 Dec).

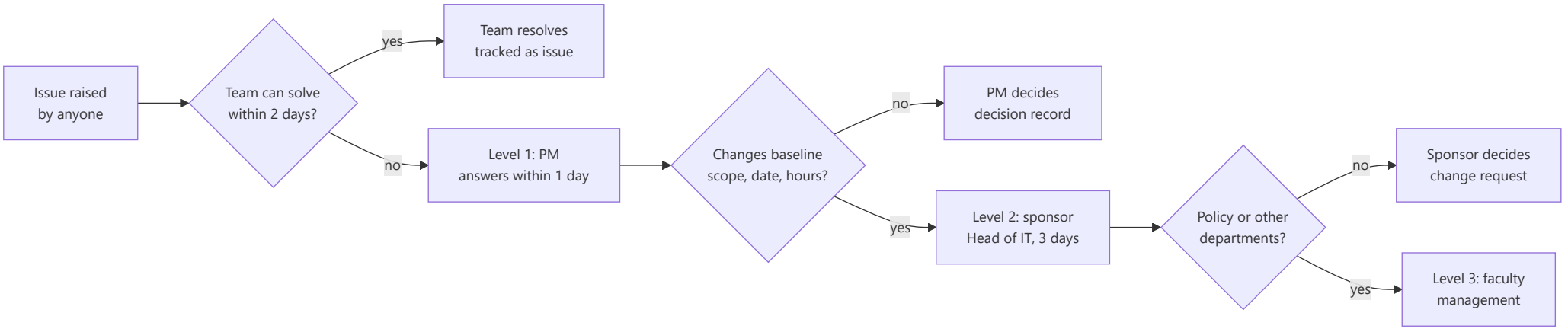
Impact: one week later; no extra cash cost; about 30
extra person-hours (EAC 736 h vs 706 h reported in week 12).
Cause: the campus proxy blocks WebSocket for 40 % of
students. The SockJS fallback works but needs a load test
with 100 simulated students, which takes 5 working days.
Plan: fallback merged 16 Dec, load test 17-21 Dec, dress
rehearsal with the 2 instructors 4 Jan, pilot 6 Jan.
What I need from you: approval of the new date by Fri 18
Dec, and a note to the instructors, which I have drafted
and attached.

I take responsibility for not testing on campus Wi-Fi
earlier; the retrospective action is in the wiki (R-03).

Best regards, Sreymom
```

⚠ Deliver bad news **early** (a risk that materialised is not a surprise if it was in the register), **in person or by call first** when the receiver is the sponsor, then in writing; **never** let the sponsor learn it from someone else.

Escalation is a defined path, not a failure



Level	Decides on	Time limit	Example
0 Team	How to do the work	2 working days	Median of 5 clock samples
1 PM	Priorities inside the baseline	1 working day	Freeze "Could" features (D-13)
2 Sponsor	Baseline: scope, date, hours, cash	3 working days	Pilot moves to week 15
3 Faculty	Policy, other departments	Next monthly meeting	Proxy rule change for all students

- Escalate **with a proposal** and the facts: "we need X by date Y, options A and B, I recommend A".
- **Conflict over information** ("the timer is fixed" vs "it is not"): first agree on the source of truth (the test on campus Wi-Fi, the tracker, the log), then separate *facts* from *interpretation*, and write the disagreement down. Most information conflicts are two people looking at two different data sets.

Information distribution: one place, one name, one class

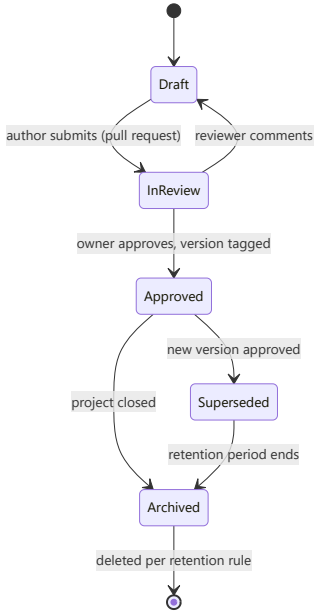
```
wiki/                          (GitHub wiki + repo docs/)
├── 00-plan/          charter.md, scope-statement.md, schedule.pod
│                     communication-management-plan.md
├── 10-decisions/     D-012-leaderboard-top-10.md
├── 20-meetings/      2026-11-20-sprint-4-review.md
├── 30-reports/       2026-11-20-status-week-08.md
├── 40-releases/      v0.4.0-release-note.md
├── 50-incidents/     2026-12-02-INC-003-ws-outage.md
└── 90-archive/       superseded versions

Naming: YYYY-MM-DD-type-topic.md or ID-topic.md
        lower-case, hyphens, no spaces, no v_final2
Owner:  every folder has one owner in the plan
Status: Draft / In review / Approved / Superseded
        in the first line of every document
```

Class	Examples (Realtime Quiz)	Where it may live	Who may read	Retention
Public	Release notes, user guide, incident notices	Public wiki, pilot channel	Anyone	Life of the product
Internal	Plan, minutes, status reports, risk register	Private repository and wiki	Team, PM, sponsor, faculty	Semester + 2 years (accreditation)
Confidential	Pilot student names and scores, feedback survey raw answers	Database, restricted folder	PM, instructors; team only anonymised	Delete raw survey data at semester end + 1 month
Restricted	JWT secret, database password, Linode token, TLS keys	Password manager, server environment only	DevOps student and PM	Rotate at hand-over and after any exposure

- **Distribution**: publish to the store, then *notify* with a link (push). Never attach the document itself: attachments fork the truth.
- **Retention** is a decision, not an accident: what is kept, how long, who deletes. Pilot students are real people; their data follows the university's data policy.
- **Least privilege**: a new team member gets Internal on day 1, Confidential when a task needs it, Restricted only in the DevOps role.

Every document has a lifecycle and an owner

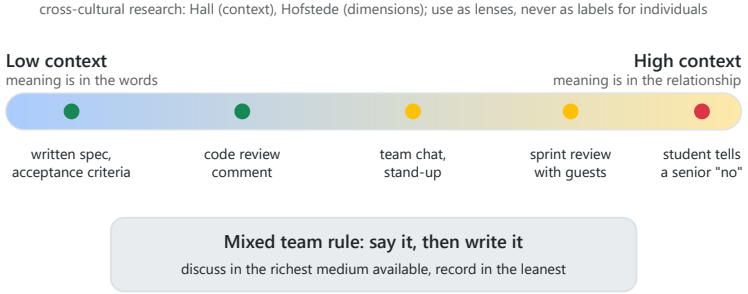


Document	Owner	Reviewer / approver	Versioning	Update trigger
Project management plan and subsidiary plans	PM	Sponsor	Git tag `plan-v1.2`, change history section	Approved change request
Status report	PM	None (published as is)	One file per week, never edited after sending	Weekly
Minutes	Rotating minute taker	Facilitator, within 24 h	One file per meeting; corrections as a dated note	Every meeting
Decision record	Person who raised it	Decision maker	Immutable; new decision supersedes	Any decision beyond level 0
Release note	Sprint presenter	PM	Matches the Git tag `v0.4.0`	Every deployment

- Git gives versioning, review (pull requests), history and access control for free: use it for documents, not only code.
- A document with no owner rots; a document with no status is trusted when it should not be. The first line of every file says both.
- Distribution lists change: when a pilot instructor joins in week 9, the plan row C3 and the wiki access are updated the same day.

Cross-cultural communication: the same words, different meanings

Dimension	What differs	In a mixed Khmer / English team	Practice
Context (Hall: high vs low)	Where meaning lives: in the words, or in relationship, tone and silence	"Yes" may mean "I heard you"; silence in a review may mean disagreement	Confirm every decision in writing; ask "what would make this hard?" instead of "any problems?"
Power distance (Hofstede)	How natural it feels to question a senior	Students may not tell the lecturer-PM that the estimate is wrong	Anonymous estimates first, then discussion; PM speaks last in planning
Individualism vs collectivism	Who gets credit and blame	Naming one person in a public retro can shame; group harmony valued	Retro talks about the system, not people; praise in public, correct in private
Time orientation	Deadlines as fixed points or as intentions	"Around Friday" is heard differently by the sponsor and the team	Dates with day and time; the calendar is the source of truth
Directness of feedback	How criticism is packaged	Direct English feedback in a code review can read as hostile	Comment on the code, not the coder; prefix with intent ("suggestion:", "blocking:")
Language	Second-language load and vocabulary	Technical English is fine, nuance and humour are not; Khmer terms for PM concepts vary	Bilingual glossary; speak Khmer, write decisions in English; no idioms in written instructions



Remote and hybrid work: a one-page team guideline

```
# Remote and cross-cultural communication guideline (team charter)
Agreed in Sprint 1 retro, 9 Oct 2026. Review every retro.

## Language
- Speak Khmer or English, whichever is faster for the speaker.
- Every decision, action and issue is WRITTEN IN ENGLISH, in full
  sentences; no idioms, no abbreviations outside the glossary.
- Glossary: wiki/00-plan/glossary.md (EN + KH), anyone may add.

## Meetings
- Camera on, microphone muted when not speaking, one speaker at
  a time; the facilitator invites the quiet people by name.
- Every meeting has a written agenda 48 h before and minutes 24 h
  after; absent members read the minutes, not the recording.
- "Any questions?" is replaced by "What would make this hard?"

## Availability
- Core hours 13:00-17:00 Mon-Fri (all on campus or online).
- Chat outside 08:00-21:00 may wait until the next morning.
- P1 incident: phone call to the on-call developer, then chat.

## Feedback
- Code review comments start with: nit / suggestion / question /
  blocking. Praise in public, correct one to one.
- Retro discusses the process and the system, never a person.
```

Remote risk	Counter-measure
Hallway information: those on campus know, those at home do not	If it matters, it goes in the tracker or the wiki the same day
Silence read as agreement	Round-robin: every participant says one sentence before a decision
Bad audio, dropped calls (noise)	Minutes written live in a shared doc; screen share, not "look at my screen"
Time zone or schedule gaps (part-time students)	Core hours; asynchronous first: written proposal, then a short call
Trust builds slowly without a face	First 5 minutes of the weekly call are not about the project

The guideline is part of the communication management plan (section 6) and is reviewed in every retrospective: rules that nobody follows are deleted, not ignored.

WRAP-UP

Best practices and common mistakes

1 Plan from the receiver's decisions

Every row of the plan names the decision or action it supports. Information nobody acts on is noise. Ask the five questions; do not guess.

2 Same format, same time, every time

Friday 16:00, seven sections, RAG with written thresholds. Predictability turns reading into a habit and makes a missing report visible.

3 Decisions and actions leave the room

Minutes within 24 hours with numbered decisions and actions (owner, due date), copied to the tracker. A meeting without them was a conversation.

✕ Mistake: green until the crash

Reports stay Green because Amber feels like failure. SPI 0.90 is Amber by the plan, so it is Amber. Early bad news is a plan; late bad news is a scandal.

✕ Mistake: everything in the chat

Decisions, credentials and 40 screenshots in one Telegram group. Chat is for now; the tracker, the wiki and the password manager are for later.

✕ Mistake: "sent" mistaken for "understood"

An email to the sponsor and a "yes" from a junior are not confirmation. Close the loop: reply expected by Monday, paraphrase, written decision.

Chapter quiz 10 questions

1. A project has 6 people who all need to talk to each other. How many communication channels exist?

- 6
- 12
- 15
- 30

2. The Realtime Quiz core team grows from 4 people to 6. How many new channels appear?

- 2
- 6
- 9
- 15

3. Week 8 status: PV = 340 h, EV = 306 h, AC = 360 h. What is the SPI?

- 0.85
- 0.90
- 1.11
- 0.94

4. BAC = 600 h and CPI = 0.85. Using EAC = BAC / CPI, what is the estimate at completion?

- 510 h
- 600 h
- 706 h
- 760 h

5. Which of these is a pull communication method?

- The Friday status email to the sponsor
- The daily stand-up
- The project wiki with the decision records
- The incident notice pushed to the pilot channel

6. In the sender-receiver model, what is "noise"?

- Only the acoustic disturbance in a video call
- Anything that distorts the message: jargon, bad audio, second language, assumptions
- The feedback the receiver sends back
- The medium chosen by the sender

7. What must the minutes of a meeting always contain?

- A full transcript of who said what
- Numbered decisions and actions with one owner and a due date
- Only the list of attendees
- The slides presented

8. How should the first lines of a bad-news email about a slipped pilot read?

- A friendly greeting and a summary of the good work done this sprint
- The headline: what changed and its impact, followed by cause, plan and ask
- An explanation of why the delay was not the team's fault
- A request to schedule a meeting to discuss

9. When should the project manager escalate an issue to the sponsor (level 2)?

- Whenever a team member disagrees with the PM
- When the issue changes the baseline (scope, date, hours, cash) and is beyond the PM's authority
- Only after the pilot has failed
- For every risk in the register

10. In a mixed Khmer / English team, which practice best addresses high-context communication and power distance?

- Hold all meetings in English only
- Ask "any problems?" at the end of each meeting
- Confirm every decision in writing and let the PM speak last when estimating
- Avoid written records so people feel free to talk

WRAP-UP

Summary

Communication succeeds only when the decoded meaning equals the intended one; feedback closes the loop, "sent" proves nothing.

Channels = n(n-1)/2: 5 people have 10, 7 have 21, 9 have 36. Route information through hubs, not private chats.

Requirements analysis: one row per stakeholder with information, why, frequency, format and sensitivity.

The communication management plan has one row per flow: audience, purpose, content, method (push, pull, interactive), frequency, owner, feedback loop.

Meetings are a system: agenda 48 h before, timebox, minutes in 24 h with numbered decisions and actions that go into the tracker.

Status = RAG with written thresholds + EVM (SPI 0.90, CPI 0.85, EAC 706 h) + top risks + decisions needed, one page, same time every week.

Write for the reader: BLUF status emails, decision records with context and options, release notes in the user's words.

Bad news: early, headline first, impact, cause, plan, ask. Escalate along a defined path, with a proposal.

Information has a home, a name, an owner, a class and a retention rule; in a mixed Khmer / English team: say it, then write it.

Open Lab-10: 5 tasks + 1 challenge

Next chapter: 11 · Stakeholder Management